

Analytics Engineer Roadmap

Goal: Transition from Power BI Lead to Analytics Engineer and eventually Analytics Architect.
Focus on owning the full lifecycle: BRD → Data Model → Pipeline → Semantic Layer → Dashboard. Learning philosophy: 70% Build, 20% Theory, 10% Courses.

Month 1: Python Foundations

Learn Git basics, Python fundamentals, Pandas, APIs and file handling. Courses: - GitHub Skills - Kaggle Python - Kaggle Pandas Build: - OTT Analytics repository - Raw → Silver pipeline - Clean, validate and standardize viewership data - Simple Power BI dashboard from Silver layer

Month 2: Analytics Engineering with dbt

Learn analytics engineering concepts and dbt. Courses: - dbt Fundamentals Learn: - Sources - Models - Tests - Documentation - Lineage Build: - Fact_Viewership - Dim_Channel - Dim_Show - Dim_Date - Dim_Region Create staging, intermediate and mart layers.

Month 3: Orchestration & Flow

Learn pipeline orchestration concepts and Airflow basics. Course: - Astronomer Airflow Learn Build: - Extract → Validate → Transform → Publish flow - Logging - Error handling - Retry strategy - Automated dashboard refresh workflow

Months 4-6

Deepen data modeling skills. Read: - The Data Warehouse Toolkit Build a complete Raw → Silver → Gold architecture. Improve semantic models and business marts.

Months 7-9

Learn Databricks and Spark concepts. Focus: - DataFrames - Delta Tables - Medallion Architecture - Optimization Convert Python ETL into Spark-based pipelines.

Months 10-12

Learn architecture and platform thinking. Read: - Fundamentals of Data Engineering Learn: - Lakehouse Architecture - Security - Governance - Cost Optimization Deliver BRD, HLD, LLD and Runbooks for projects.

Priority Skills

1. SQL 2. Data Modeling 3. Business Understanding 4. Python 5. dbt 6. Databricks 7. Airflow 8. Architecture Do not focus on Scala initially. Focus on understanding systems and trade-offs.